

The Goalkeeper in Possession

Visible Options, Retention Risk, and Asymmetric Turnover Cost

Author information omitted for blind review

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Abstract

Introduction. Goalkeeper distribution is usually evaluated after the pass is played, yet coaches decide before it: build short, play through pressure, or go direct. We ask whether the options visible to a goalkeeper explain those choices and improve retention modeling, and how that probability should be interpreted when failure near goal is disproportionately costly.

Methods. We analyze 45,974 goalkeeper distributions from 722 StatsBomb open-data matches across five elite cohorts. A target-free expected-retention model (xR-GK) is evaluated with true leave-one-competition-out testing and independent out-of-fold value models. We additionally analyze 4,831 goalkeeper distributions with StatsBomb 360 freeze frames from the 2022 World Cup and Euro 2024. Visible teammates are classified by distance, centrality, receiver pressure, and whether a 1.5-m passing corridor is clear. On 4,802 exact joins to frozen xR-GK, five-fold match-grouped models test whether two 360 variables - clear short-option count and a visible central short exit - add information beyond locally recalibrated xR, event pressure, goal-kick state, and competition.

Results. Direct play was observed on 85.9% and 91.9% of distributions with no clear short exit in the World Cup and Euro samples, respectively, falling to 27.2% and 25.8% with two or more clear short exits; the pattern survived completed-pass and lane-width sensitivities. Adding the two 360 variables improved match-held-out AUC from 0.648 to 0.699 (delta=0.051, 95% CI 0.035-0.066), Brier score by 0.0065, and log loss by 0.0167, with the same direction in both tournaments and after visible-area control. Event-only xR retained moderate discrimination in unseen competitions (AUC 0.645-0.710) but absolute probabilities shifted materially. Independently refit cohort models also showed that opponent consequence after failure exceeded the origin possession value in every environment (C/V 1.40-1.90), making the complete lost branch 2.40-2.90 times origin value.

Conclusion. Goalkeeper distribution is an option-constrained decision under asymmetric risk. The visible option set explains both short-versus-direct behavior and retention beyond event-only controls, while probability scale remains environment-dependent and failure near goal is systematically costly. For clubs, the practical task is not to assign one universal “good with his feet” score, but to ask what options the goalkeeper sees, how likely each distribution is to retain possession in the relevant environment, and what the team risks if it fails.

1. Introduction

The goalkeeper receives the ball and the coaching question arrives before the pass: can we play short, can we play through or around the press, or should we go over it? For opposition analysis the question reverses: what short exits can we remove, and where is the goalkeeper likely to play when those exits disappear? A useful goalkeeper-distribution model should therefore describe the decision state the goalkeeper faces, not only the pass observed after the decision.

“Good with his feet” is related but different. Technical comfort includes receiving, first touch, back-pass handling, one-touch play, disguise and use of both feet. Distribution adds choice: which teammate or space is selected, under what pressure, with what probability of retaining possession, and at what consequence if the ball is lost. Recruitment adds a third question - fit - because the same distribution does not carry the same absolute probability in every league, team structure or pressure environment.

This paper develops that hierarchy in three layers. First, xR-GK estimates the ex-ante probability that a goalkeeper distribution reaches a retained state. Second, a two-branch value formulation prices what is gained if the action is retained and what opponent state is created if it fails. Third, StatsBomb 360 freeze frames expose part of the option set visible when the goalkeeper acts. The tracking layer allows a direct test of a limitation of event-only models: they may know the origin, pressure and action context but not whether a clean teammate was actually available.

The paper makes five empirical contributions. (1) In two tournaments, observed direct play falls sharply as the number of visible lane-clear short exits rises. (2) Two fixed option-set variables add held-out retention information beyond locally recalibrated frozen xR and non-360 controls. (3) Event-only xR discriminates moderately across unseen competitions but its absolute probability level shifts. (4) Independently refit value models show an asymmetric lost branch in every cohort. (5) Intended destination geometry strongly improves retention discrimination when observable, while its observation process is itself selected.

2. Related work and novelty

The study sits at the intersection of goalkeeper decision analysis, passing risk, possession value and tactical option availability. Lamas et al. separated goalkeeper decision quality conceptually from realized action outcome, while Yam's Sloan goalkeeper framework combined shot stopping, crosses, defensive activity and distribution. Power et al. modeled pass risk and reward with tracking data; Anzer and Bauer modeled pass difficulty with contextual spatiotemporal features; VAEP and expected-possession-value work formalized the broader problem of valuing soccer actions. Recent work has also emphasized that expected-threat models require explicit validation before scouting use.

Goalkeeper distribution is increasingly studied as behavior rather than isolated completion. Ding et al. analyzed 29,911 WSL goalkeeper passes using graph-based clustering and showed that spatial context, defensive pressure and possession phase organize passing behavior. FIFA's goalkeeper language - around, through, into, onto and beyond - likewise frames distribution as tactical intention rather than a homogeneous pass type.

Our novelty is narrower than "first goalkeeper model." It is the combination of: an event-data probability model whose portability is tested out of competition; an explicit asymmetric lost branch; a target-free/target-aware identification design; and an event-linked 360 experiment showing that visible short-option structure both explains observed short/direct choice and adds held-out retention information beyond non-360 controls. The objective is an evidence hierarchy for what public data can and cannot identify, not a universal goalkeeper leaderboard.

3. Football questions and analytical framework

3.1 Coach questions

- Our goalkeeper: against this opponent, should we build short, play through a central exit, or go direct?
- Opposition goalkeeper: when we press and remove the obvious short routes, where does the distribution shift?
- Recruitment: is the keeper technically comfortable, does the keeper distribute well, and does that distribution profile fit the actions our game model will require?
- Training: which recurring match states should the goalkeeper unit rehearse this week?

3.2 Retention probability

Let $R(a)=1$ denote retained resolution of goalkeeper distribution a at the first controlled team action or explicitly retained restart. The target-free model estimates $\rho(a)=P[R(a)=1 \mid X_{\text{ex-ante}}(a)]$. xR-GK is therefore an action-risk probability rather than a pure passing-skill rating. Teammate spacing, receiver quality and option availability remain embedded in the observed outcome.

3.3 Two-branch value

Let $V(s)$ be the origin possession value, $S(a)$ the retained-branch payoff, and $C(a)$ the expected opponent consequence conditional on failure. The canonical score is $xT\text{-GK}(s,a)=\rho(a)S(a)-V(s)-[1-\rho(a)]C(a)$. The origin state is surrendered in either branch. We retain the xT-GK label for continuity, but the quantity is better interpreted as an expected net distribution value than conventional attacking xT. The analysis later shows that the conceptual lost branch matters more robustly than fine action-specific estimation of $C(a)$.

3.4 Visible option set

StatsBomb 360 is event-linked freeze-frame data, not continuous tracking. For each covered goalkeeper distribution we treat visible teammates as candidate visible options, never assuming off-camera players were unavailable. We define a visible lane-clear short option as a teammate within 25 m whose goalkeeper-to-receiver segment contains no visible opponent within a preregistered 1.5 m corridor. A visible central short exit adds location in the central pitch third. These are spatial opportunity descriptors, not causal labels of the "correct" pass.

Figure 1. Goalkeeper distribution as an option-constrained decision under asymmetric risk

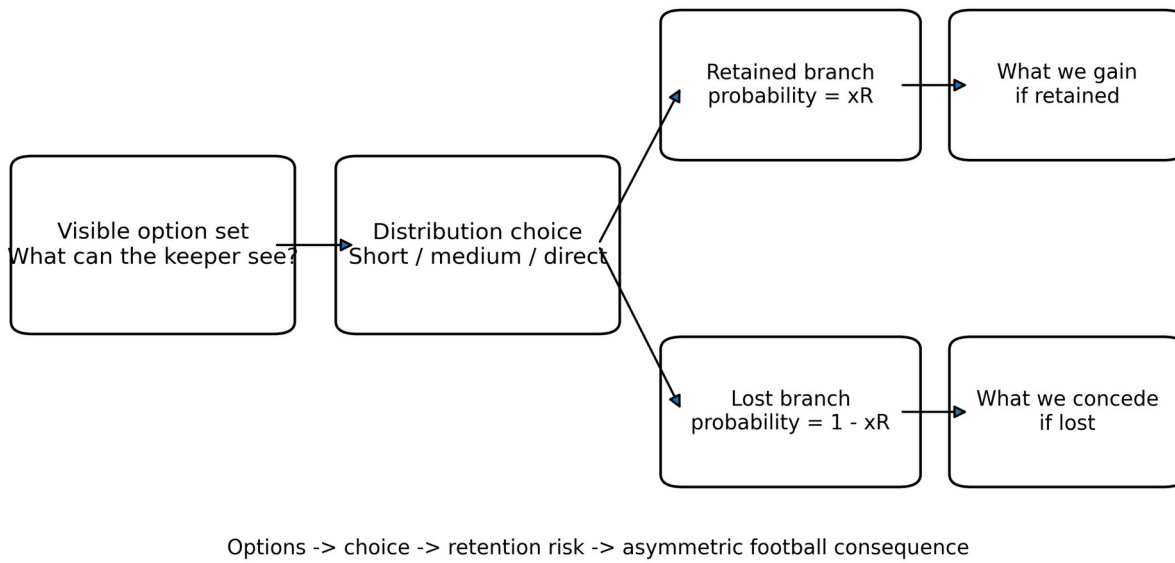


Figure 1. The analytical framework: visible options constrain the distribution choice; xR -GK estimates the retained branch; the value layer prices asymmetric retained and lost outcomes.

4. Data

The event-data study uses 45,974 eligible goalkeeper distributions from 722 StatsBomb open-data matches across Premier League 2015/16, FA WSL 2020/21, World Cups 2018 and 2022, and Euro/Copa America 2024. The target-identifiable sample contains 40,154 distributions. The 360 extension uses official Hudl/StatsBomb open data from 64 World Cup 2022 and 51 Euro 2024 matches. It contains 4,831 goalkeeper distributions with usable event-linked 360 frames; 4,802 join exactly by event UUID to frozen event-only xR predictions.

Layer	Coverage	Matches	GK distributions	Primary purpose
Event study	5 cohorts	722	45,974	Probability portability, value, target-aware analysis
360 extension	WC2022 + Euro2024	115	4,831	Visible option set and choice
360 + frozen xR join	WC2022 + Euro2024	115	4,802	Incremental retention information

The 360 sample is selected rather than random. Covered and uncovered goalkeeper distributions differ substantially in goal-kick and event-pressure prevalence. All 360 claims are therefore explicitly conditioned on the event-linked 360 sample. Freeze frames do not reveal pre-event movement, and off-camera options are unobserved rather than unavailable.

5. Methods

5.1 Event-only xR and portability

The target-free xR model uses only pre-action features such as origin position, event pressure, goal-kick state, match time, score state and body part; realized endpoints and downstream outcomes are prohibited as predictors. True leave-one-competition-out retraining excludes the target competition from training and global calibration. Discrimination, Brier score, calibration slope and mean prediction gap assess transportability.

5.2 Value and failure asymmetry

Canonical origin-value and success/failure components are fit separately within each cohort using match-grouped folds and scored out of fold. In the primary ≤ 35 m region we compare mean opponent consequence C with origin

possession value V and report C/V and the full lost-branch burden $(V+C)/V$. Ten-thousand match-cluster bootstrap draws retain repeated match multiplicity.

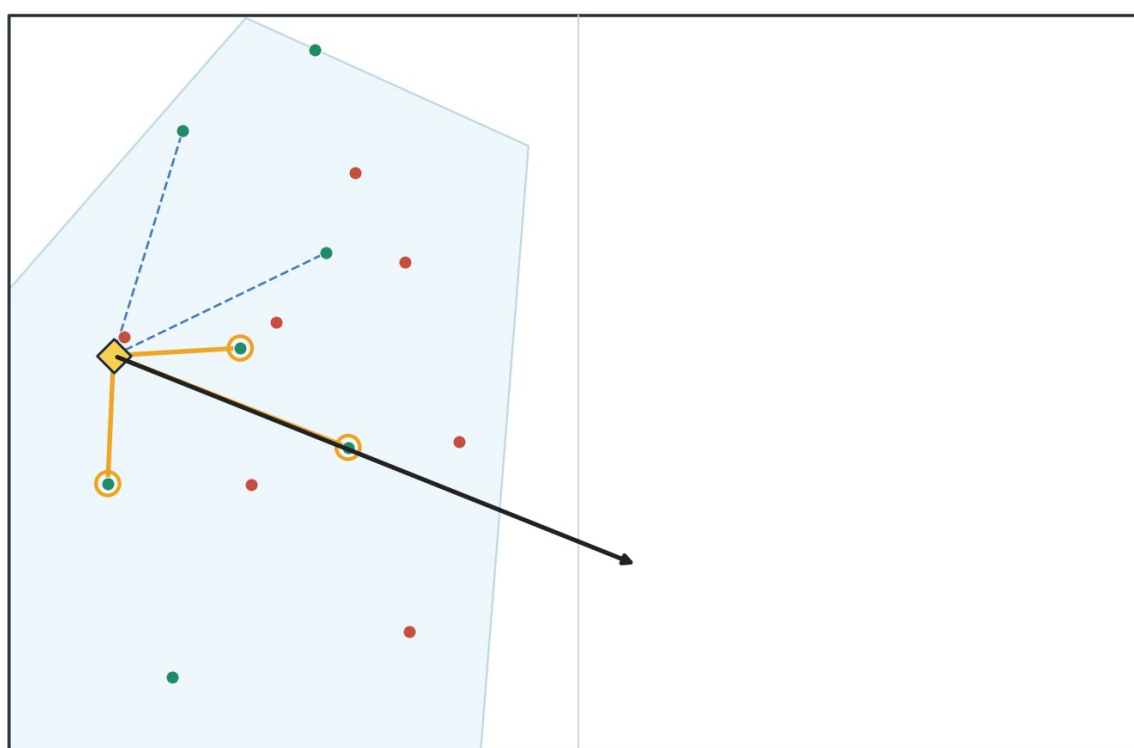
5.3 Target-aware geometry

A secondary TA-GEO model adds intended-target geometry on actions where the target can be reconstructed defensibly. Because target identification is strongly outcome-dependent, TA-GEO is a selected-sample comparison and never an all-action population model.

5.4 360 visible-option geometry

For every 360-covered goalkeeper distribution, visible teammates are characterized by distance, side, nearest visible opponent and lane obstruction. Distance families are short <25 m, medium 25-40 m and direct ≥ 40 m. The primary lane corridor is 1.5 m, with 1.0 m and 2.0 m sensitivities. The option-choice analysis compares observed short/medium/direct distributions at 0, 1 and 2+ lane-clear short exits, separately by tournament and in completed-pass sensitivity.

EURO2024 · match 3930175 · event 001f36d5-65a3-4a1b-8fb5-fc48dbf14a90
Observed DIRECT · clear short 3 · central exit YES · pressure 1



Green: visible teammate · red: visible opponent · gold diamond: GK · black: observed pass
Solid blue/gold: visible 1.5m lane-clear short candidate · dashed blue: lane blocked · gold ring: central short exit
Provider recipient field present; endpoint arrow does not establish reception.

Figure 2. Example StatsBomb 360 decision state. Green points are visible teammates, red points visible opponents, solid gold lines lane-clear short candidates, dashed lines blocked short candidates, the ring marks a visible central short exit, and the black arrow is the observed pass. Off-camera players are unobserved, not unavailable.

5.5 Does 360 add information beyond event-only xR?

The incremental-information experiment uses identical five-fold GroupKFold splits by match. M2 is the primary non-360 control: logit(frozen xR), event pressure, goal-kick state and competition. M3 adds exactly two 360 terms: lane-clear short-option count and visible central short exit. M3 and M2 are compared on identical held-out actions with AUC, Brier score and log loss. Five-thousand paired match-cluster bootstrap draws provide uncertainty. A robustness model additionally controls for the fraction of the pitch visible in the 360 frame.

5.6 Criterion validation and falsification

Separately from the 360 experiment, the event-only framework is evaluated against downstream signed shot-xG outcomes. These analyses do not support treating full xT-GK as a superior predictive goalkeeper metric: xR is the more stable ex-ante criterion predictor, and action-specific C(a) does not clearly outperform fixed or coarse failure penalties. These negative results constrain the interpretation of the value layer.

6. Results

6.1 The visible option set strongly structures short versus direct choice

In both tournaments, direct distribution decreases sharply as visible clear short exits increase. In World Cup 2022, direct play falls from 85.9% with zero clear short exits to 43.0% with one and 27.2% with two or more. Euro 2024 shows the same pattern: 91.9%, 57.1% and 25.8%. Each additional lane-clear short option is associated with lower direct-play odds in match-clustered logistic models (OR 0.64 in World Cup 2022 and 0.60 in Euro 2024). A visible central short exit also predicts lower direct odds. Event pressure moves in the opposite direction.

The finding is not an artifact of failed-pass endpoints. Among completed passes only, direct play falls 72.9% → 22.7% → 12.2% in World Cup 2022 and 80.2% → 32.0% → 11.0% in Euro 2024 across 0, 1 and 2+ clear short exits. The same direction survives 1.0 m, 1.5 m and 2.0 m lane-corridor definitions.

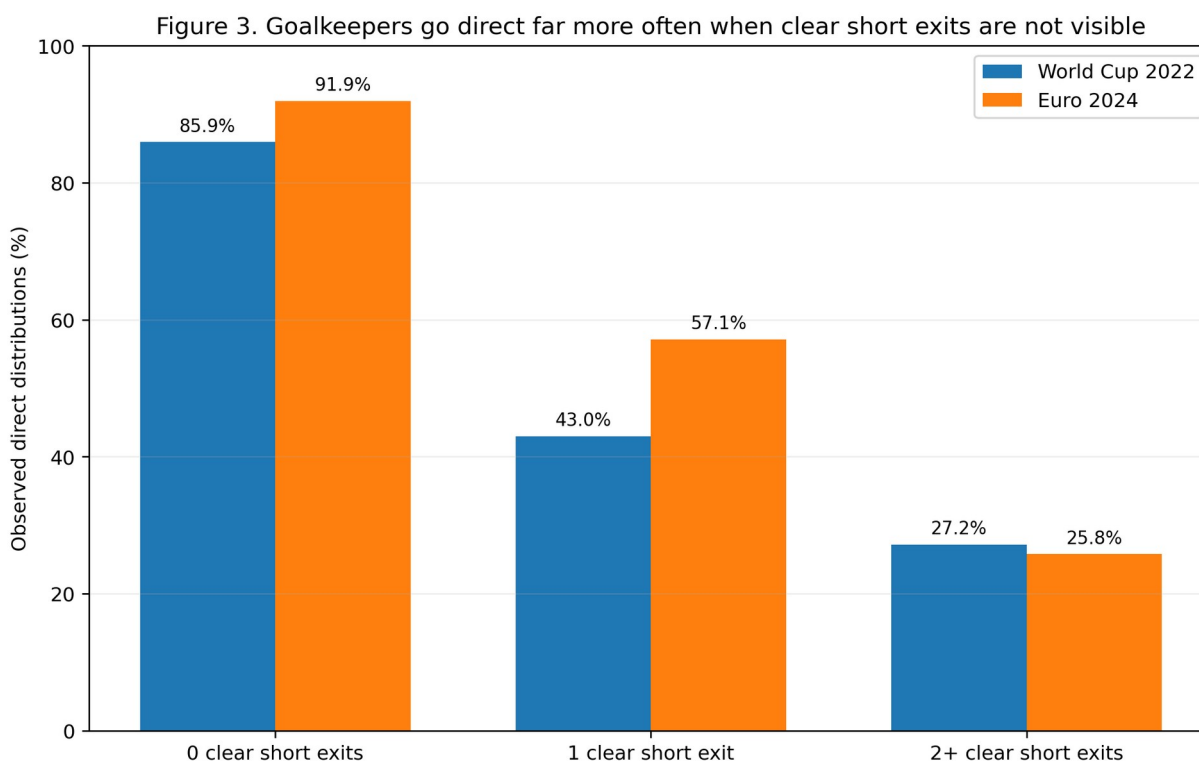


Figure 3. Observed direct distribution by number of visible lane-clear short exits. The relationship is strong and directionally consistent in both tournaments.

6.2 Visible-option information adds retention information beyond event-only controls

On 4,802 exact UUID joins, M3 improves match-held-out retention prediction over M2 on all three metrics. Pooled AUC rises from 0.6485 to 0.6991 (delta +0.0506, 95% CI +0.0348 to +0.0660); Brier score falls from 0.1661 to 0.1596 (delta -0.0065, 95% CI -0.0087 to -0.0042); and log loss falls from 0.5046 to 0.4879 (delta -0.0167, 95% CI -0.0221 to -0.0111). The improving direction holds separately in both tournaments. Adding visible-area fraction as a camera-coverage control does not remove the result.

Ablation shows that clear-short count provides the larger single pooled contribution (delta AUC +0.0481), while central exit alone also adds information (+0.0414). The variables are correlated and should not be interpreted causally or as separable tactical effects.

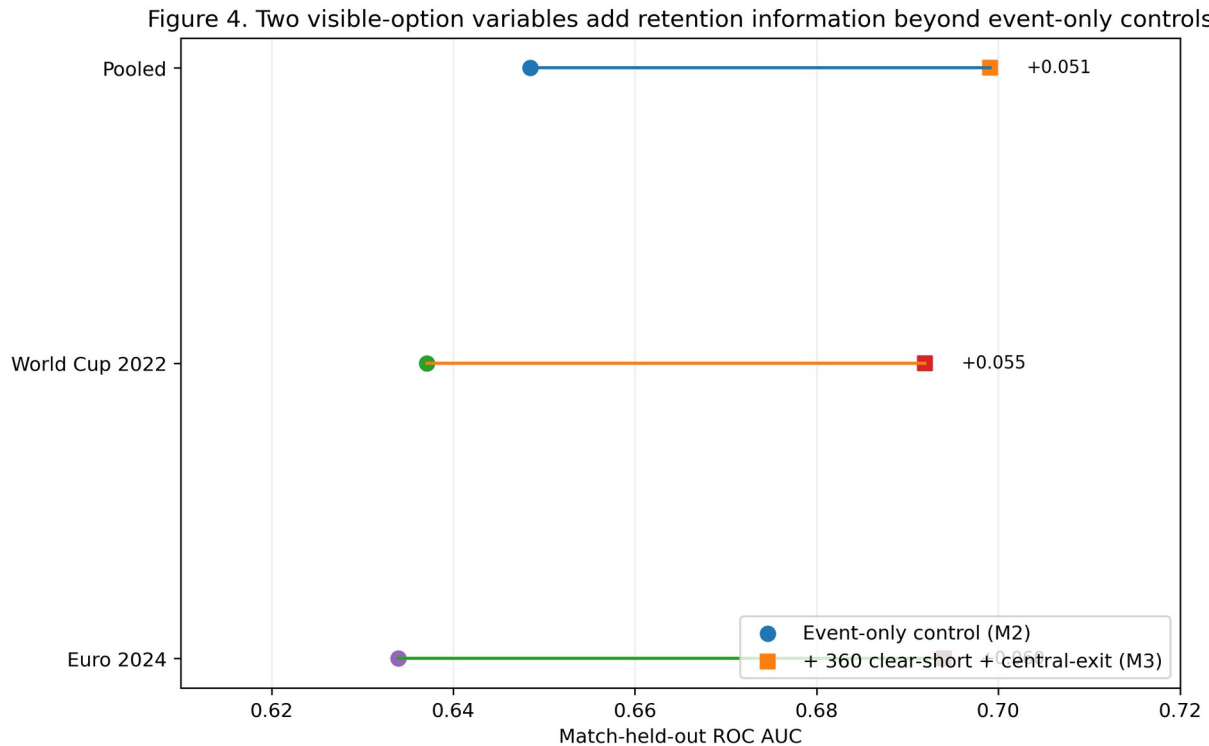


Figure 4. Two fixed 360 option variables add match-held-out retention information beyond locally recalibrated frozen xR, event pressure, goal-kick state and competition. The direction holds in both tournaments and after visible-area control.

6.3 Event-only xR misses part of the option state

The frozen xR residual pattern is consistent with omitted option-state information. When no lane-clear short exit is visible, event-only xR overpredicts retention by about 26 percentage points in World Cup 2022 and 21 points at Euro 2024. With two or more clear short exits, it slightly underpredicts retention in both competitions. This is a diagnostic rather than causal proof, but together with the held-out augmentation test it shows that visible option structure contains information absent from the event-only state.

6.4 Retention probability transports imperfectly across football environments

True leave-one-competition-out evaluation produces moderate unseen-competition discrimination (AUC 0.645-0.710). Absolute probability scale shifts materially: mean predicted-minus-observed retention ranges from +7.4 to -9.5 percentage points, and calibration slopes range from 0.67 to 1.46. Relative risk structure therefore travels better than absolute probability, motivating environment-aware calibration in recruitment and analysis.

6.5 Failure consequences are asymmetric

Independent cohort-specific out-of-fold refits produce heterogeneous magnitudes but the same directional result in every environment. Opponent consequence C is 1.40-1.90 times origin possession value V, making the full lost branch (V+C) 2.40-2.90 times the value of the state being surrendered. The asymmetry is greatest closest to the goalkeeper's own goal and generally declines outward.

Cohort	Opponent consequence C/V	95% CI	Full lost branch (V+C)/V
PL 2015/16	1.40x	1.39-1.40	2.40x
Euro+Copa 2024	1.47x	1.45-1.48	2.47x
WC 2018	1.90x	1.86-1.95	2.90x
WC 2022	1.47x	1.45-1.50	2.47x
WSL 2020/21	1.74x	1.71-1.76	2.74x

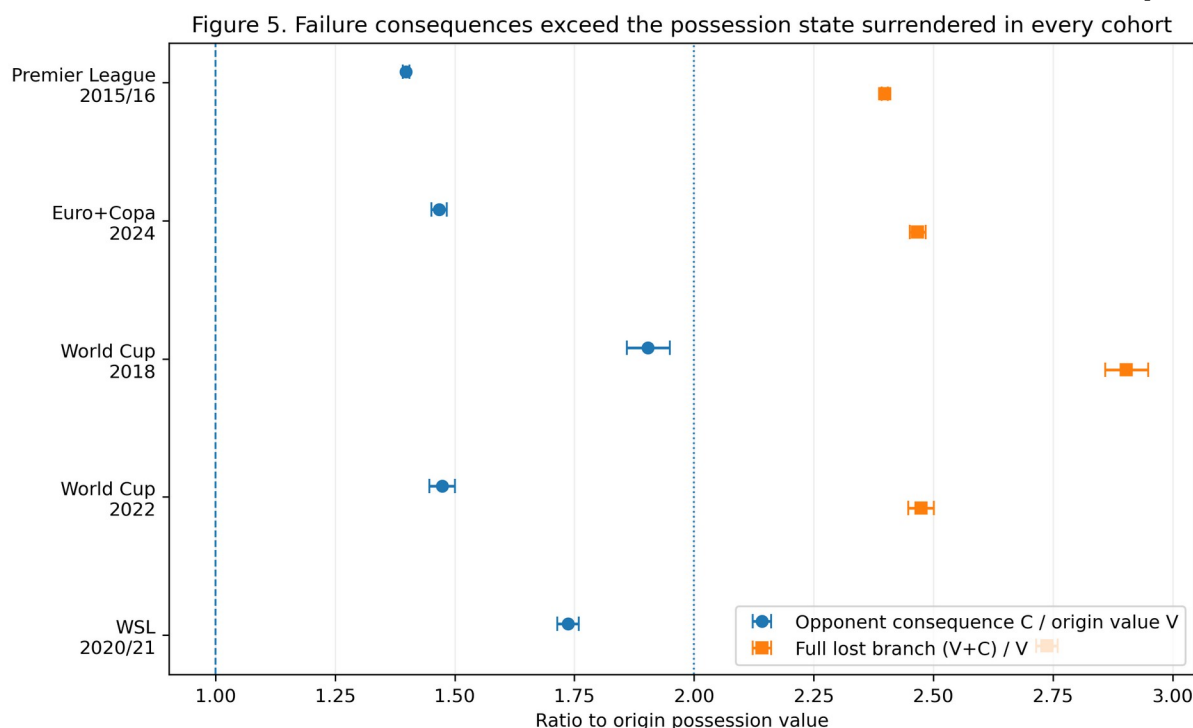


Figure 5. Independently refit out-of-fold failure consequence exceeds origin possession value in every cohort. The full lost branch includes both the surrendered origin state and opponent consequence.

6.6 Intended destination remains highly informative, but selected

On 40,154 identical target-identifiable actions, adding intended geometry improves AUC from 0.659 to 0.849 overall (delta +0.189). By cohort, gains range from +0.151 to +0.244. However, target identification is far more complete on retained than failed distributions, so the result is selected-sample evidence rather than an all-action population estimate.

6.7 Full xT-GK is not the hero predictive metric

Downstream criterion validation provides an important negative result. xR is the most stable tested ex-ante predictor of later signed shot-xG outcomes. Failure pricing improves the weak success-only construction in some settings, but full action-specific $C(a)$ is statistically indistinguishable from fixed or coarse failure penalties in all paired keeper-level comparisons. Team-adjusted analyses likewise favor xR more consistently than full xT-GK. We therefore use the value layer as a decision-theoretic consequence scale, not as a claimed universal predictive ranking.

7. Application: from metric to match plan

The practical unit is not one goalkeeper score. It is a decision profile.

7.1 Our goalkeeper: should we play short or go direct?

The 360 result provides a direct starting point: the visible state of short exits is strongly associated with whether the goalkeeper stays short or goes direct. Staff can combine that option map with the team's build-up principles: if two clean short exits are visible, the historic data show much less direct play; when no clear short exit is visible, direct play dominates. The analysis does not prescribe the correct action, but it identifies the picture that should trigger the coaching decision.

7.2 Opposition goalkeeper: what happens when we squeeze the short exits?

Event-data scouting profiles can now be written in coach language. In the 2024 sample, Jordan Pickford's observable distribution shifts from 30.9% direct when free to 56.0% direct under the event pressure flag. Historical Kasper Schmeichel is already highly direct in both states (79.6% free; 83.8% pressured), while Grace Moloney remains

predominantly short (61.6% free; 57.4% pressured). These are descriptive patterns to take to video and training, not causal claims that a particular press will reproduce the same response.

7.3 Recruitment: good with feet, good distributor, good for us

These should be separated. “Good with feet” includes technique not fully identified by public event data. “Good distributor” combines action selection, difficulty, retention and pressure response. “Good for us” asks whether those decisions remain appropriate in the receiving club’s game model. Because probability calibration changes across environments, raw completion or xR should not be transferred mechanically from one competition to another.

7.4 Training

The same framework converts directly into training pictures: play around the first line when the wide exit is clear; play through the central exit when the lane exists; escape over the press when short exits are closed; and rehearse the second-ball structure that makes the direct option part of the plan. The analytical output should end in repeatable game states, not in a dashboard.

8. Robustness, falsification, and limitations

- StatsBomb 360 is event-linked freeze-frame data, not continuous tracking. It does not observe the checking movement that created an option, and off-camera players are unobserved rather than unavailable.
- 360 coverage is selected: covered and uncovered goalkeeper distributions differ in goal-kick and event-pressure prevalence. The option-set findings describe the covered sample from WC2022 and Euro2024.
- The option-choice associations are observational. Removing a short passing lane may not causally produce the same distribution response in a different match or team.
- Event-only xR is not pure individual skill. Team structure, receiver quality and game model contribute to outcomes.
- Intended-target observation is outcome-dependent, so TA-GEO is selected-sample evidence.
- Full xT-GK is not a superior downstream predictor to xR, and action-specific failure cost does not clearly outperform simpler penalties.
- Earlier transported C/V magnitudes around 2.2, completion-inversion, style-neutrality, near-perfect zero-shot calibration and a universal 40-60 m restart penalty were rejected or downgraded during validation.

9. Conclusion

Goalkeeper distribution is best understood as an option-constrained decision under asymmetric risk. The visible option set is not a cosmetic addition: in two international tournaments, it strongly structures short-versus-direct behavior and adds held-out retention information beyond locally recalibrated event-only xR and non-360 controls. The probability branch itself remains environment-dependent, while independent value models show that failure near goal creates an opponent state more costly than the possession state surrendered. The practical implication is not one universal “good with his feet” number. Clubs should ask what options the goalkeeper can see, how difficult the available distributions are in the relevant environment, and what the team gains or risks through each branch of the decision.

Data, code, and reproducibility

The open-source submission repository should contain frozen match manifests, official StatsBomb/Hudl source URLs and hashes, canonical event-model code, exact 360 option definitions, model-selection locks, environment files, processed publication inputs, figure scripts, and reproduction instructions. The 360 provenance audit freshly downloaded and validated 348 official JSON files from a pinned Hudl open-data commit and reproduced all 4,831 distributions and 27,777 candidate options without unexplained scientific-field discrepancies. Repository URL: <INSERT PUBLIC REPOSITORY URL BEFORE SUBMISSION>.

AI-assisted research disclosure

Generative AI-assisted tools were used for research planning, code generation/review, robustness-test design, literature discovery, document drafting and editorial revision. The human author retained responsibility for the scientific definitions, analyses, interpretation, citations and final manuscript. Numerical claims are sourced to

reproducible code and evidence files; no AI system is listed as an author and no synthetic research observations were used.

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